Computer Networks Interview Guide

Q1. What is a computer network?

A **computer network is a group of connected devices that communicate with each other and share data or resources**.
Example: When your laptop connects to Wi-Fi and accesses a website, your laptop communicates with servers over a network.

Q2. What is the difference between LAN and WAN?

LAN (Local Area Network) covers a relatively small area, such as a home, office, or college.
WAN (Wide Area Network) connects networks across large geographical areas, such as cities or countries.
Example: Home Wi-Fi → LAN; Internet → WAN.

Q3. What is an IP address?

An **IP address is a logical address assigned to a network interface/device so it can be identified and communicated with over an IP network**.
Versions: IPv4 (32-bit), IPv6 (128-bit).
Example: 192.168.1.10. It is like a **postal address** for network communication.

Q4. What is the difference between IPv4 and IPv6?

IPv4 uses **32-bit addresses** (e.g., 192.168.1.10), while **IPv6** uses **128-bit addresses** (e.g., 2001:db8::1) to provide more unique addresses.

Q5. What is a MAC address?

A **MAC address is a 48-bit hexadecimal link-layer address associated with a network interface**, used for local network communication.
Example: 00:1A:2B:3C:4D:5E. It acts as a permanent identifier for your hardware.

Q6. What is the difference between MAC address and IP address?

MAC Address	IP Address
Link-layer address	Network-layer address
Local network delivery	Routing across networks
48 bits	32 bits (v4) / 128 bits (v6)

Q7. What is a port?

A **port is a logical number (0-65535) used to identify a particular application or service on a device**.
Example: HTTP (80), HTTPS (443), SSH (22). IP identifies the machine; Port identifies the service.

Q8. What is a socket?

A **socket is a communication endpoint defined by an IP address + Port**.
Example: 192.168.1.10:443. IP is the building address; port is the specific apartment.

Q9. Explain the OSI model.

The **OSI model is a 7-layer conceptual model**: 7. Application, 6. Presentation, 5. Session, 4. Transport, 3. Network, 2. Data Link, 1. Physical.

Q10. What happens at each OSI layer?

7. Application: Services like HTTP, DNS. **6. Presentation:** Encoding/Encryption. **5. Session:** Dialogue management. **4. Transport:** End-to-end delivery (TCP/UDP). **3. Network:** IP Routing. **2. Data Link:** MAC frames. **1. Physical:** Raw bits/signals.

Q11. What is the difference between OSI and TCP/IP?

OSI is a 7-layer conceptual model. **TCP/IP** is a practical 4-layer model (Application, Transport, Internet, Network Access) used for the actual Internet.

Q12. Which layer does a router work at?

A router primarily operates at **Layer 3 (Network layer)** using IP addresses for forwarding.

Q13. Which layer does a switch work at?

A traditional switch operates at **Layer 2 (Data Link layer)** using MAC addresses.

Q14. Which layer does HTTP work at?

HTTP is an **Application-layer protocol (Layer 7)**.

Q15. What is the difference between TCP and UDP?

TCP	UDP
Connection-oriented, Reliable	Connectionless, Best-effort
Ordered Byte Stream	Datagram oriented

Q16. Why is TCP reliable?

TCP ensures reliability via **sequence numbers, acknowledgments, retransmissions, and flow/congestion control**.

Q17. What is the TCP 3-way handshake?

Establishes a connection via: **1. SYN** (Request), **2. SYN-ACK** (Response), **3. ACK** (Acknowledgment).

Q18. What happens during TCP connection termination?

Uses a **4-step exchange (FIN, ACK, FIN, ACK)** as each side independently closes its direction.

Q19. What is flow control?

Prevents a fast sender from overwhelming a slow **receiver** using a receive window.

Q20. What is congestion control?

Prevents overwhelming the **network** by adjusting sending rates when congestion is detected.

Q21. Flow control vs congestion control?

Flow control: Protects the receiver. **Congestion control:** Protects the network.

Q22. What is a TCP connection?

A logical, stateful relationship between two endpoints for reliable, ordered data exchange.

Q23. What happens when you type google.com?

Flow: 1. DNS Lookup (get IP) → 2. TCP connection → 3. TLS Handshake → 4. HTTP Request → 5. Server Response → 6. Browser Renders HTML/CSS/JS.

Q24. What is the difference between HTTP and HTTPS?

HTTPS is HTTP over **TLS**, adding encryption and authentication. HTTP uses Port 80; HTTPS uses Port 443.

Q25. What is the difference between GET and POST?

GET: Retrieves data. **POST:** Submits data for processing or creation.

Q26. What is the difference between PUT and PATCH?

PUT: Replaces the full resource. **PATCH:** Partially modifies the resource.

Q27. What are HTTP status codes?

Categories: 2xx (Success), 3xx (Redirection), 4xx (Client Error - e.g., 404), 5xx (Server Error - e.g., 500).

Q28. What is REST?

An architectural style using HTTP methods (GET, POST, etc.) to manage **resources** represented by URLs.

Q29. What is an API?

An **Application Programming Interface** allows software to interact via a defined contract (e.g., Request → Waiter → Kitchen).

Q30. What is DNS?

The **Domain Name System** translates domain names (google.com) into IP addresses, like a phone contacts list.

Q31. What is DHCP?

Automatically assigns IP addresses, masks, and gateways to devices joining a network.